GEK-73869A
Supersedes GEK- 73869

Instructions

One-Cell Thermal Gas Analyzer Equipment

3S7576HA101

User's Manual

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GEK-73869A

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These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia, U.S.A.

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Safety Symbol Legend

WARNING

Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION

Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

NOTE

Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

WARNING

This equipment contains a potential hazard of electric shock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and the instructions should install, operate, or maintain this equipment.

Isolation of test equipment from the equipment under test presents potential electrical hazards. If the test equipment cannot be grounded to the equipment under test, the test equipment's case must be shielded to prevent contact by personnel.

To minimize hazard of electrical shock or burn, approved grounding practices and procedures must be strictly followed.

WARNING

To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained persons should modify any programmable machine.

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CHAPTER 1

OVERVIEW

1-1. INTRODUCTION

These instructions cover the installation, operation, and adjustment of the 3S7576HA101 Thermal Gas Analyzer equipment. This equipment is designed to monitor and indicate hydrogen purity in a simplified hydrogen-cooled generator with a vacuum-type sealing system.

1-2. EQUIPMENT OVERVIEW

The equipment consists of the following parts:

- power supply panel
- detector panel
- separately-mounted cell block
- purity indicator
- filter-dryer unit
- milliammeter
- balanced-bridged current adjusting rheostat

The purity indicator gives a continuous reading of hydrogen purity by indicating percent hydrogen in air over a range of 50% to 100%. It includes high and low alarm set-pointers, adjustable to any point on the scale. The purity indicator is calibrated for use when a second or remote purity indicator is used. The second or remote purity indicator is generally located in the control room and is not equipped with alarm contacts.

The filter-dryer eliminates foreign particles and moisture from the gas.

The milliammeter and bridge current rheostat are used for calibration.

1-3. POWER AND MOUNTING REQUIREMENTS

The equipment operates on a 120 volt, 50/60 Hz single-phase power source.

WARNING

Hazard of electrical shock or burn. Only personnel that are adequately trained and are thoroughly familiar with the equipment and these instructions should install, operate, and maintain this equipment.

The thermal gas analyzers are structurally designed for floor-mounted, conventionally ventilated indoor and outdoor type enclosures.

1-4. RECEIVING AND HANDLING

1-4.1. Receiving and Handling

If assistance is needed, contact:

GE Drive Systems
Product Service Engineering
1501 Roanoke Boulevard
Salem, Virginia 24153, U.S.A.
(703) 387-7595

1-5. STORAGE

If the equipment is not installed immediately upon receipt, it must be stored properly to prevent corrosion and deterioration. Since packing cases do not protect the equipment for outdoor storage, the customer must provide a clean, dry storage area free of temperature variations, high humidity, and dust.

Use the following guidelines when storing the equipment:

1. Place the equipment under adequate cover with the following requirements:

   a. Keep the equipment clean, dry, and protected from precipitation and flooding.

b. Use only breathable (canvas type) covering material - do not use plastic

2. Unpack the equipment as described in section 1-6 and label it.

3. Maintain the following environment in the storage enclosure:

   a. Ambient storage temperature limits from -20 °C (-4 °F) to 55 °C (131 °F).

   b. Surrounding air free of dust and corrosive elements, such as salt spray or chemical and electrically conductive contaminants.

   c. Ambient relative humidity from 5 to 95% with provisions to prevent condensation.

   d. No rodents.

   e. No temperature variations that are severe enough to cause moisture condensation on the equipment.

CAUTION

Moisture on certain internal parts can cause electrical failure.

Condensation occurs with temperature drops of 15 °C (27 °F) at 50% humidity over a 4-hour period, and with smaller temperatures variations at higher humidity.

If the storage area temperature varies in such a way, install a reliable heating system that keeps the equipment temperature slightly above that of the ambient air. A 100-watt lamp can sometimes serve as a substitute source of heat.

CAUTION

To prevent fire hazard, remove all cartons and other such flammable materials packed inside units before energizing any heaters.

If equipment is to be stored for an extended period of time, keep it heated to about 5 °C above ambient temperature to keep out moisture.

1-6. UNPACKING

If the equipment has been exposed to low temperatures for an extended period of time, allow the equipment to reach room temperature before unpacking it.

When unpacking, check the contents of each case against the packing list. Report any shortage to GE Drive Systems.

Use standard unpacking tools, including a nail puller. Carefully remove the equipment from its container to avoid damaging or marring the part. Wipe off any particles of packing materials or foreign substances that may be lodged in or between the parts.

Small parts (such as bolts and screws are packed in special containers to keep them together. However, they may become separated. Therefore, carefully inspect packing material for loose parts before discarding it.

CHAPTER 2

DESCRIPTION OF OPERATION

2-1. THERMAL CONDUCTIVITY THEORY

The thermal conductivity of a gas is a measure of its ability to conduct heat. It is defined as the "time rate of transfer of heat by conduction through a unit thickness of gas across a unit area for a unit difference in temperature". In the metric system, it is measured in calories per second per square centimeter for a difference in temperature of 1 °C.

In thermal conductivity work, the relative conductivity (the ratio of conductivity of a gas to the conductivity of air) is often used. In relative conductivity, two gases, helium and hydrogen, stand apart having conduction about six to seven times as great as air.

The high thermal conductivity of hydrogen makes it possible to run large hydrogen-cooled generators at higher loads for a given temperature rise because more rapid removal of heat from the winding and iron structure is possible.

High thermal conductivity is also an excellent indicator of hydrogen purity because a mixture of any gas with hydrogen registers lower thermal conductivity than hydrogen gas. About 10 per cent helium in hydrogen has about the same thermal conductivity as one per cent air in hydrogen. (Helium has about 90 per cent of the conductivity of hydrogen.) This is not a power station problem, since there are no ordinary sources of helium to contaminate hydrogen in the usual system.

All other gases have about the same thermal conductivity as air, although air is the only contaminant usually present. Analyzers are calibrated for per cent of hydrogen in air.

2-2. HARDWARE DESCRIPTION

The 3S7576HA101 Thermal Gas Analyzer equipment is composed of the following units, unless otherwise noted:

- A 3S7576SA200 semi-enclosed type power supply panel containing a voltage transformer, a full-wave silicon rectifier bridge, a zener voltage regulator, fuses, and an outgoing connection terminal board.
- A 3S7576SA202 semi-enclosed detector panel containing bridge output and balance rheostats, resistors, and outgoing connection terminal board.
- A 3S7576SA103 test cell block containing a cell block, and isolation bushings at the gas inlet and outlet ports. The cell block is used in conjunction with a 3S7576SA202C detector panel. These cell blocks are normally mounted in a separate gas section of a control cubicle.
- A 44A370834-001 (type 195 meter-relay) purity indicator, including hermetically sealed plug-in relays and sockets, one high set point, and one low set point. Mounted on the studs of the meter and in parallel with the meter element is a 105 ohm precision resistor that may be removed when parallel operation of a remote purity indicator is required.
- A 933B579P2 (D091) dc milliammeter (with a red line at the 325 milliamp mark).
- A potentiometer per drawing 44A370848-D10 (10 ohm, 2W).
- Miscellaneous hardware for mounting the above components in the floor-mounted enclosures.
- A 3S7576SA107 filter assembly containing a purifier and cartridge, a moisture and impurity indicator, and fittings. This unit is intended for use with generators rated up to 90 psi hydrogen gas.

Figure 2-1 shows the primary parts of the thermal gas analyzer equipment. The power supply receives 120 V ac. A gas sample is introduced into the gas analyzer cell block, which electrically connects to the detector panel.

Bridge current is adjusted by the bridge current rheostat and indicated by the bridge-current milliammeter. The type 195 (meter-relay) purity indicator shows the per cent hydrogen in the air. The indicator's adjustable high and low alarmset-pointers initiate the externally-mounted control unit relays of the type 195 (meter-relay) purity indicator at predetermined set points. (Some units may be equipped with a remote purity indicator.)

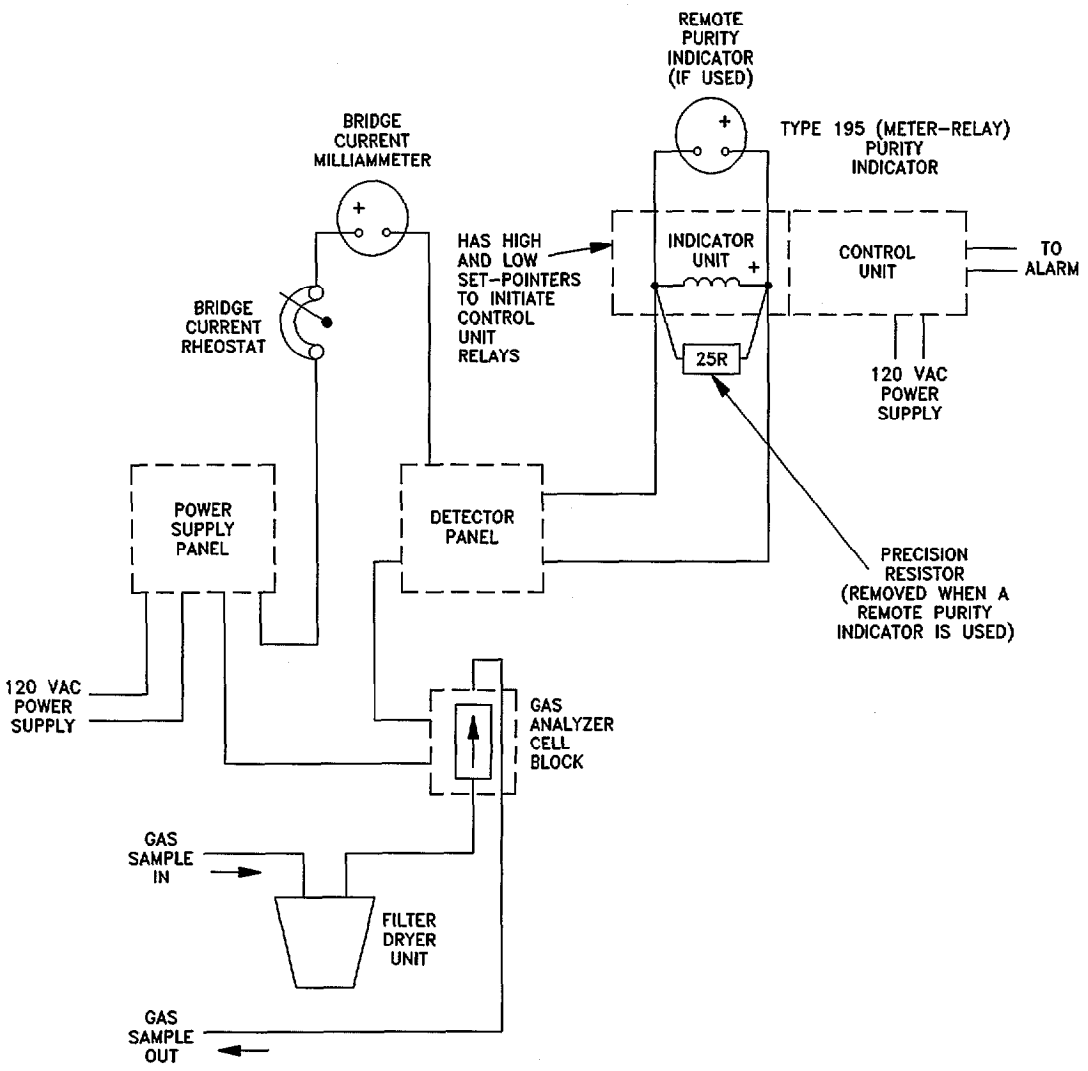


Figure 2-1. Simplified Block Diagram of Electrical Equipment

2-2.1. Piping Circuit

Figure 2-2 shows a simplified piping circuit for the thermal gas analyzer equipment. The gas analyzer cell block receives gas from the generator. The gas flow is regulated by opening the throttling valve until the desired flow is obtained. Flow should be regulated as explained under Section 4-6.

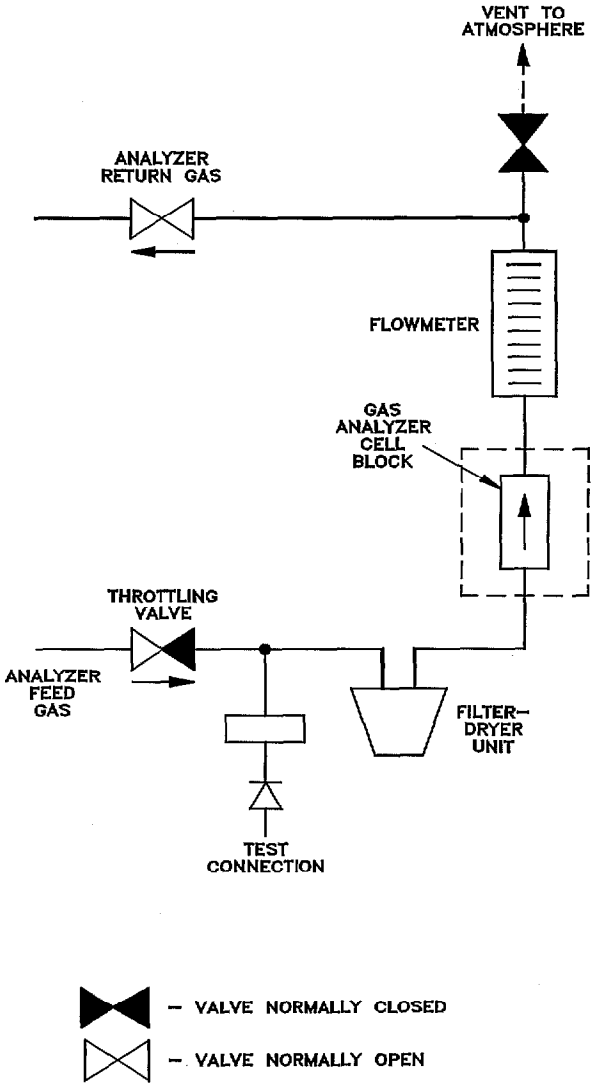


Figure 2-2. Simplified Piping Circuit

2-2.2. Gas Analysis Filaments

Figure 2-3 illustrates the construction of a typical gas analyzer cell block. A fine helical wire filament is placed axially in each of two cylindrical chambers in the metal cell block. One end of each filament is mechanically supported by the filament support structure and is thereby electrically connected to the main section of the cell block. Since the cell block is electrically insulated from ground potential, only one end of the filament is insulated from the main section of the cell block.

Provision is made in the cell block for the hydrogen gas to be admitted to the hydrogen test chamber. Both filaments are heated by an electric current. Some of the heat is lost by conduction to the filament supports and by radiation and convection. However, most of the heat is transferred from the filament to the cell structure by conduction through the gas surrounding the filament.

The temperature and resistance of each filament is then an inverse function of the thermal conductivity of the gas surrounding the filament. The thermal conductivity of the hydrogen increases with its purity. The temperature and resistance of the filament in the hydrogen chamber therefore decrease as the hydrogen purity of the gas surrounding the filament increases. Therefore, if the filament is used as one leg in a wheatstone bridge, the change in resistance of the filament as the gas purity changes can be used to indicate the hydrogen purity of a sample gas passed through the hydrogen test cell chamber containing the hydrogen filament.

The helium reference filament maintains essentially constant resistance, since the filament is surrounded by a sealed helium gas.

Variations in ambient temperature and other heating effects are approximately canceled since both filaments are mounted in the same metal enclosure.

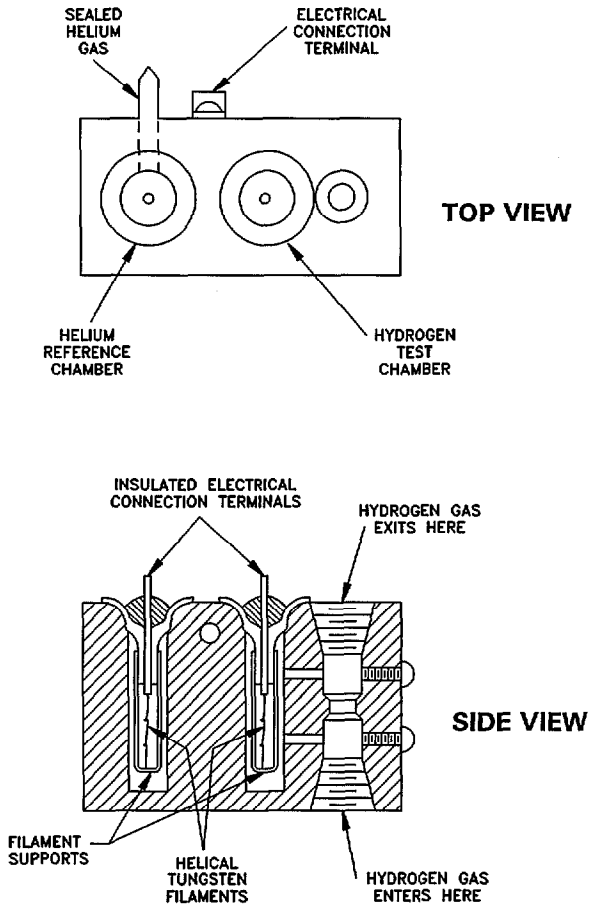


Figure 2-3. Typical Gas Analyzer Cell Block Construction

2-2.3. Electrical Circuit

Figure 2- 4 illustrates the electrical circuit of the thermal gas analyzer equipment. A 120 V ac supply is connected through fuses 1FU and 2FU to voltage transformer 1T. The output of 1T is rectified by a full-wave SCR bridge 1REC. The output voltage of this bridge is dropped across resistor 1R and regulated by zener 1Z.

The regulated voltage is the dc supply for the wheatstone bridge circuit. The essential components of the bridge circuit are a H_2 analyzer filament, an He reference filament, bridge resistors 2R, 3R, 4R, and bridge balance rheostat 2P.

If bridge balance potentiometer 2P is adjusted so that the series-parallel combination of resistors 3R and 4R equals the resistance of 2R, the bridge will be in balance if the resistance of the H_2 analyzer filament equals the resistance of the He reference filament.

This means that no potential appears across the output terminals of the bridge AB. If the H_2 analyzer filament resistance decreases, a positive voltage appears at terminal B. A current then flows from terminal B through the purity indicator 63-4X/K-292 and the bridge output rheostat 3P to terminal A of the bridge circuit. The more the H_2 analyzer filament resistance decreases, the more current flows through the purity indicator.

The bridge circuit is normally balanced at the factory so that no current flows through the purity indicator when a mixture of 50% hydrogen and 50% nitrogen gas is passing through the analyzer cell. With this condition, the output of the bridge is zero and the purity indicator reads 50% hydrogen in air. This corresponds to zero current through the purity indicator. Bridge balance potentiometer 2P is used to make a fine adjustment of this point. As the purity of the gas increases toward 100% hydrogen, the resistance of the hydrogen analyzer filament decreases and current flows through the purity indicator from plus to minus.

When the test gas is 100% hydrogen, the purity indicator reads 100% hydrogen in air. This corresponds to 1 milliamp through the purity indicator. Bridge output potentiometer 3P is used to make a fine adjustment of this point. The bridge current flowing through milliammeter M-1 is 325 milliamps.

The exact value of the bridge current is not critical, but the current must not change after the gas analyzer is calibrated. This increase in current increases the sensitivity of the purity indicator to changes in hydrogen purity, and causes the scale readings to expand outward from the center of the scale. Readings on the lower end of the scale decrease and readings on the upper end increase. Readings in the region of 70 to 80 per cent hydrogen in air are affected only slightly. A decrease in bridge current causes the upper end of the scale to read low and the lower end of the scale to read high.

2-2.4. Remote Purity Indicators

All purity indicators are calibrated for use with a remote purity indicator. If a remote purity indicator is required for indication of the generator-hydrogen purity at a location remote from the gas analyzer equipment, remove resistor 25R connected in parallel with the (type 195 meter-relay) purity indicator.

2-2.5. Filter-Dryer Unit

The filter-dryer unit uses a replaceable desiccant purifier cartridge shipped in a hermetically- sealed can. This cartridge must be replaced if the drying agent becomes saturated by moisture.

The moisture and impurity indicator indicates the presence of moisture in the gas line and the resulting exhaustion of the desiccant in the purifier. The indicating element is filled with silica gel. The normal blue color of this desiccant agent turns to pink when used.

The expended desiccant agent may be regenerated by heating the indicating element in an oven at 150 °C for two or three hours. A manufacturer's knurled-cap plug assembly, which contains the desiccant, is available as a replacement for this unit. It is usually possible to keep the generator hydrogen at a low-relative humidity so that frequent replacement of the desiccant materials is not necessary.

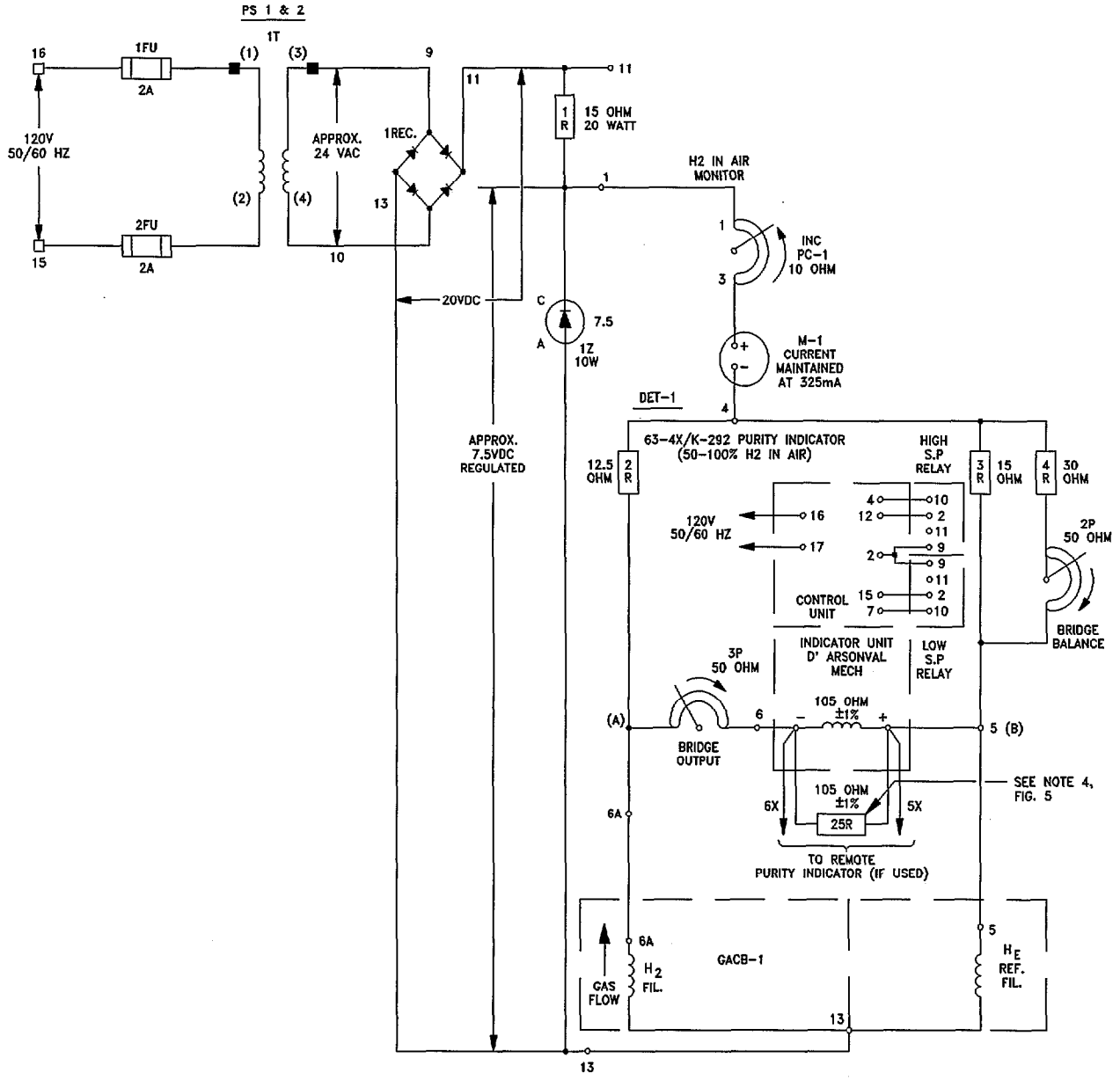


Figure 2-4. Electrical Circuit

CHAPTER 3

INSTALLATION

3-1. LOCATION AND MOUNTING

The various components of the thermal gas analyzer equipment may be supplied unmounted or panel-mounted within a floor-mounted hydrogen control cubicle. It is important that all components be placed in a well ventilated, clean, dry location and securely mounted on surfaces that are free from vibration.

The instruments and bridge-current rheostat unit must be semi-flush mounted. All other components are arranged for surface mounting.

WARNING

Under certain conditions, gas will explode if ignited. To prevent explosion, and possible injury to personnel, the instructions below must be strictly followed.

Pure hydrogen is normally nonflammable. However, a mixture between 5 and 75 per cent hydrogen in air will explode if ignited in an enclosed space. To ensure complete safety of operation of the gas analyzer equipment:

- Maintain a hydrogen purity in the detector block between 97 and 98 percent hydrogen in air at all times, thus making an explosion normally impossible.
- Mount the gas analyzer equipment in a properly ventilated control cabinet in which all electrical controls that might cause an arc are either behind a gas-tight seal or are hermetically sealed-type switches. This precaution greatly reduces the possibility of a fire from any leak in the gas piping, which could collect and build up to an explosive mixture.

3-2. CONNECTIONS

Connections must be made as shown in diagram supplied with the equipment.

CAUTION

To avoid equipment damage, make sure all connections are correct.

Notes:

CHAPTER 4

OPERATION AND ADJUSTMENTS

4-1. OPERATION

The following steps are required to place the equipment in operation:

1. Regulate the flow as described section 4-6.

2. Adjust potentiometer PC-1 so that the bridge-current is at the red line mark (325MA) on the bridge-current milliammeter M-1.

3. Take readings of the gas purity from the type 195 (meter-relay) purity indicator.

4. The high and low alarm set-pointers on the type 195 (meter-relay) purity indicator may be set to any point by adjusting the two knobs located on the front of the instrument.

The gas analyzer detector panel requires no attention during operation, except to check flow rate and bridge-current immediately before taking readings. Large changes in gas concentration cause minor changes in bridge-current. Ordinarily, voltage fluctuations on a 120 volt power supply does not cause current variations because of the regulated voltage across the zener in the 3S7576SA200B panel.

4-2. THERMAL GAS ANALYZER CALIBRATION

If a component such as a purity indicator or a cell block is replaced, the thermal gas analyzer equipment may need recalibrating. This can be accomplished by either of two methods described below. The method used depends on the component being replaced.

4-2.1. Approximate Recalibration Method

This method is used after a minor fault has been corrected. An example of a minor fault is a sticking needle on the purity indicator.

Recalibration is accomplished by purging the test cell block with a commercially pure hydrogen gas and adjusting potentiometer 3P until the purity indicator reads 99.6%. This method of calibration eliminates any error at the upper end of the scale.

Any error existing at the 50 per cent point is reduced proportionally as scale readings increase, becoming zero at the upper end of the scale. For example, an error at the 50 percent point would be reduced to one-tenth of the error when the pointer reads 95 percent hydrogen in air.

The purity indicator is accurate only when the bridge current is set to the red line of the milliammeter scale (325MA). This is particularly true at low-scale deflection. At the center and high end of the scale, readings appear independent of the bridge-current.

The speed at which the purity indicator reads the gas purity depends on the capacity of the line feeding the instrument. If the gas runs rapidly through the instruments, disregard the readings during the high-flow. When the flow is reduced to normal, the purity indicator reads the correct value of the gas within a few seconds.

Water vapor causes some changes in the reading. Machine gas is usually dry enough so that effects of water vapor are negligible. Free water in the gas analyzer cell should be avoided because it changes the calibration, and water contaminated with oil and other substances may permanently damage the cell.

4-2.2. Exact Recalibration Method

This method of calibration is used after a major fault has been corrected. An example of a major fault is the failure of the cell block.

The following procedure is recommended for recalibration:

1. Study Figure 2-3, which shows a typical gas analyzer cell block construction and the locations of the H_2 and H_e helical tungsten filaments. These filaments are also shown in the electrical circuit, Figure 2-4.

2. Make sure that a tank of 50 to 55% hydrogen-nitrogen gas be available. A 200-cubic foot tank makes about 30 calibrations, sufficient for years of service.

3. Calibrate the gas analyzer equipment by balancing the wheatstone bridge, points A and B, to zero output with the 50% hydrogen, and maximum output (full scale) with 100% hydrogen in the analyzer cell.

4. In practice, it is usually not possible to obtain an exact 50% hydrogen-nitrogen mixture. In this event, all zero output adjustments should be made at the percent hydrogen labeled on the tank. That is, for a 52% hydrogen-nitrogen gas mixture, all zero output adjustments should be made at the 52% mark on the purity indicator scale.

5. To simplify the calibration procedures, make all adjustments assuming use of a 50% hydrogen-nitrogen mixture. All circuit references are made using the electrical circuit, Figure 2-4.

4-3. TESTING SEQUENCE

1. Install the gas analyzer cell block in place of the defective unit.

2. Adjust the sliders of potentiometers 2P and 3P, located in the detector panel in the mid-position.

3. Regulate the flow of the 50% hydrogen-nitrogen gas, in the cell block, as described in section 4-4.

4. When the cell block is charged with the 50% hydrogen mixture:

   a. Adjust PC-1 until the milliammeter in the bridge circuit reads 325MA (red line).

   b. Adjust potentiometer 2P until the needle on the purity indicator reads at the 50% mark.

   c. With the calibration complete for the 50% point on the purity indicator, purge the cell block with 100% hydrogen gas.

5. When the cell block is charged with the 100% hydrogen gas, adjust the milliammeter in the bridge circuit to read 325MA (red line). The adjustment of the current should always be made when switching from 50 to 100% hydrogen gas or vice versa.

6. Adjust potentiometer 3P (clockwise), located in the detector panel, until the needle of the purity indicator reads at the 100% mark.

7. To ensure accuracy of adjustment, repeat steps 2 thru 6 using the 50 and 100% hydrogen gas until there are two repetitive ratings.

4-4. FLOW CONTROL

A flowmeter is used to indicate flow through the gas analyzer cell block (see Figure 2-2). Normally, the hydrogen flow is regulated by adjusting the separate throttling valve until approximately 1.0 STD cubic foot per hour or 470 cubic centimeters per minute, at 0.5 PSIG, at 70 °F of pure hydrogen gas, is maintained on the Flowrator.

4-5. FLOW ERROR

Flow should always be adjusted below the level at which flow error becomes apparent. Flow error can be determined by the purity indicator readings. If the flow-rate is above normal, the purity indicator tends to read on the high side. For example: With a 50% hydrogen gas in the system and an abnormal flow-rate, the purity indicator reads approximately 52 to 57% instead of 50% H_2-IN-AIR.

Sensitivity to flow error also depends on the conductivity of the gas, the error being less as the purity of the hydrogen increases. Individual cells may vary slightly in their sensitivity to flow. In any case, the maximum permissible flow may be determined by increasing flow until flow error is evident.

Sufficient time should be allowed to scavenge the pipe lines with the gas to be measured before readings are taken. High flow rate may be used to scavenge the lines, but readings should not be taken until the flow is adjusted below the limits of flow error.

4-6. REMOTE PURITY INDICATOR

The remote purity indicator must be adjusted to full scale while it is in parallel with the type 195 (meter-relay) purity indicator. One milliamp at 105 millivolts from the thermal gas analyzer bridge is used to adjust the remote purity indicator with the type 195 (meter-relay) purity indicator. This may be obtained from the gas analyzer by filling the cell with pure hydrogen gas and adjusting the bridge-current to 325 milliamperes.

Maintain the reading of the type 195 (meter-relay) purity indicator at 100% by adjusting the bridge-current output potentiometer 3P. Adjust the remote purity indicator to read 100% hydrogen in air by adjusting the rheostat located in the remote purity indicator.

Notes:

CHAPTER 5

MAINTENANCE AND TROUBLESHOOTING

5-1. MAINTENANCE

Routine maintenance is limited to changing the used activated alumina in the filter dryer units.

5-2. TROUBLESHOOTING

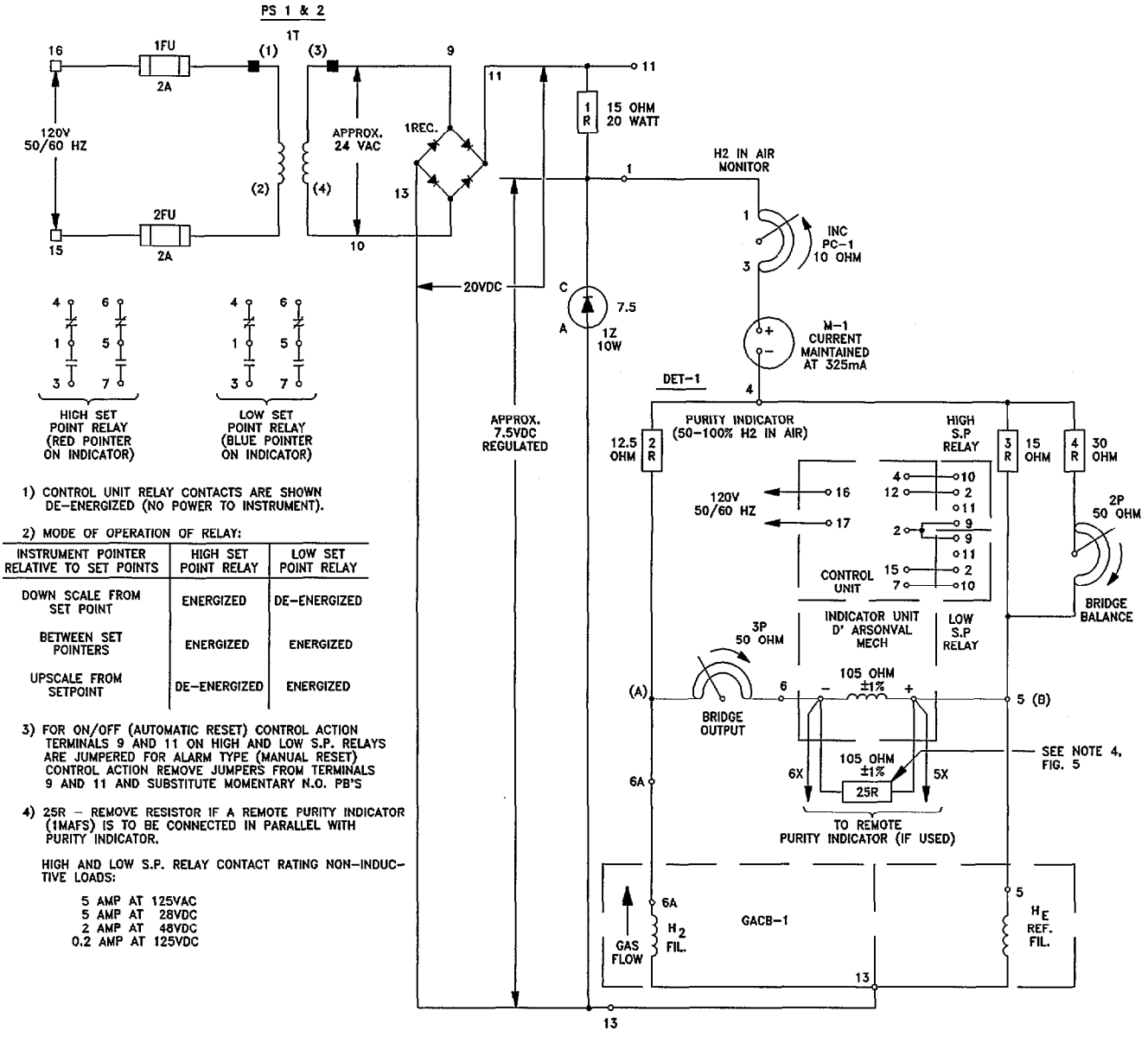
If the thermal gas analyzer equipment operates abnormally, immediately remove the equipment from service. Then carefully test the equipment to determine the cause of difficulty, using the instructions in the previous chapters.

If any connections change, fully test the affected circuits using the applicable section of these instructions. Then return the equipment to service.

Figures 5-1 and 5-2 show the elementary (electrical) and connection/interconnection diagrams for the Gas Analyzer.

Table 5-1. Troubleshooting Chart

Trouble	Probable Cause and Remedy
Purity Indicator moves off-scale to the right	Shorting or cooling of the cell filament by water, oil or other foreign matter. Check for source of trouble; when cleared, replace cell block Sealed helium reference gas has leaked out. Replace cell block.
Bridge-current tends to decrease and may reach the point where it is no longer possible to set current to red line (325MA) on milliammeter.	Sealed helium reference gas has leaked out. Replace cell block.
Bridge current increases.	Shorted cell block filaments. Replace cell block.
Irregular movement of the needle in the purity indicator when the balance-bridge potentiometer 2P is adjusted.	Broken wires on 2P. Replace 2P. Irregular contact resistance on 2P caused by dust and lint. Replace 2P.
Discontinuity or jump of needle in purity indicator.	Broken wires on 2P. Replace 2P.
Purity indicator drops to its mechanical zero.	Broken wires on 2P or 3P (when slider contact crosses over broken wire). Replace 2P or 3P.
Purity indicator readings become unsteady and then shift, indicating an apparent increase in hydrogen purity.	Flow has increased beyond a certain amount. Shut off flow for short time, gas analyzer will immediately assume correct reading.



1) CONTROL UNIT RELAY CONTACTS ARE SHOWN DE-ENERGIZED (NO POWER TO INSTRUMENT).

2) MODE OF OPERATION OF RELAY:

INSTRUMENT POINTER RELATIVE TO SET POINTS	HIGH SET POINT RELAY	LOW SET POINT RELAY
DOWN SCALE FROM SET POINT	ENERGIZED	DE-ENERGIZED
BETWEEN SET POINTERS	ENERGIZED	ENERGIZED
UPSCALE FROM SETPOINT	DE-ENERGIZED	ENERGIZED

3) FOR ON/OFF (AUTOMATIC RESET) CONTROL ACTION TERMINALS 9 AND 11 ON HIGH AND LOW S.P. RELAYS ARE JUMPERED FOR ALARM TYPE (MANUAL RESET) CONTROL ACTION REMOVE JUMPERS FROM TERMINALS 9 AND 11 AND SUBSTITUTE MOMENTARY N.O. PB'S

4) 25R – REMOVE RESISTOR IF A REMOTE PURITY INDICATOR (1MAFS) IS TO BE CONNECTED IN PARALLEL WITH PURITY INDICATOR.

HIGH AND LOW S.P. RELAY CONTACT RATING NON-INDUCTIVE LOADS:

5 AMP AT 125VAC
5 AMP AT 28VDC
2 AMP AT 48VDC
0.2 AMP AT 125VDC

Figure 5-1. Elementary Diagram for Gas Analyzer

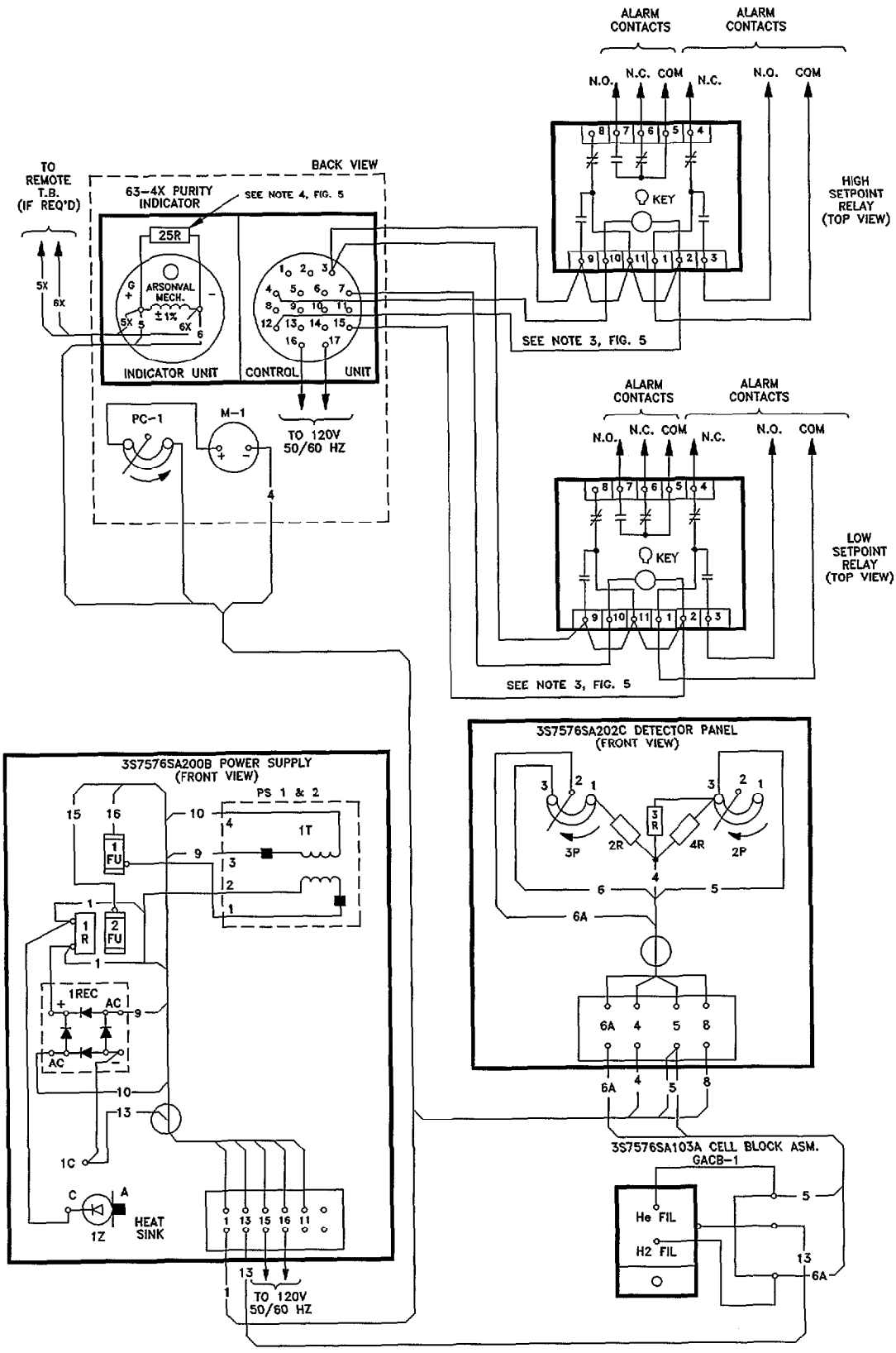


Figure 5-3. Connection and Interconnection Diagram for Gas Analyzer

Notes:



GE Drive Systems

General Electric Company
1501 Roanoke Boulevard
Salem, Virginia 24153

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